

Title: Single bin QPAD (SQPAD) approach for robust analysis of point count data with detection error

Authors: Subhash R. Lele¹ and Péter Sólymos²

¹ Department of Mathematical and Statistical Sciences, University of Alberta and Analythium Solutions Inc., Edmonton, Alberta. (slele@ualberta.ca)

² Department of Biological Sciences, University of Alberta and Analythium Solutions Inc., Edmonton, Alberta. (solyms@ualberta.ca)

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ABSTRACT

Bird population monitoring is often conducted using point count surveys. Accounting for detection errors is a major challenge in analyzing these data. The commonly used methods for correcting detection errors in counts of organisms are Distance sampling, Removal sampling, N-mixture, and QPAD. These methods rely on multiple surveys or subdivisions within surveys (time/distance bins). The reliability of these approaches depends on the accurate estimation of distance, correct identification of individuals, and the closed population assumption. Errors in distance estimation, double counting, and mortality and migration of individuals within and between survey periods can lead to substantial biases in population density estimation. Furthermore, tracking individuals and estimating distances can be difficult in field conditions. We propose a simple modification of the QPAD method so that field observers are required to collect information only about either the occupancy status or count of individuals within a specified time interval and a specified spatial buffer, a “single bin”, around the observer’s location. We show that population density parameters are identifiable by changing the time interval and the radius of the spatial buffer for each survey location. We show that this variable-effort survey method is robust against errors in distance estimation and double counting. We also show that data collected under current protocols in North America can be analyzed using SQPAD. We illustrate our methodology with biologically realistic simulations and a reanalysis of some field data.

Keywords: Catch per unit effort, Closed population, Distance sampling, Effective Detection Radius, Identifiability, N-mixture, QPAD, Removal sampling, RSPF condition.

LAY SUMMARY

- Not all birds that are present will be detected by an observer. Hence the number of birds recorded during a survey is likely to be smaller than the number that were present.
- Current methods to account for detection errors require accurately tracking individuals and their distances from the observer. This information is difficult to collect in the field.
- Double counting and errors in distance estimation can lead to biased conclusions about bird populations. These errors are exacerbated when several species are being monitored simultaneously.
- Alternative methods that do not require individual identification and distance to the observer rely on multiple visits to the same location that are logistically and financially demanding.
- We propose a new method called Single Bin QPAD (SQPAD) that accounts for detection error without requiring information on the exact distance to the observer, individual identification or multiple visits. This approach simplifies the field logistics for bird monitoring programs.
- We demonstrate that SQPAD is a valid estimation method that is also robust against double counting of individuals and incorrect distance estimation.

INTRODUCTION

Landbird monitoring surveys generally collect data about either the occupancy status or count of individuals of a species at various spatial locations in the study area (Blondel et al. 1981). These

data are used to estimate the population density for the species of interest (Edwards et al. 2023). However, even when an individual is present, it may not always be detected (Thompson 2002). Left uncorrected, such detection errors lead to underestimation of population density. For a discussion of sources of detection errors and whether one should correct for such detection errors, see Johnson (2008). We believe it is essential to account for detection errors for justifiable inferences about the status of the populations.

Detection error can be thought of as ‘visibility bias’. It is reasonable to assume that the probability of detecting an individual depends on how far that individual is from the observer, the closer the individual, the higher the probability of detection. If the distance from the observer is known, this ‘visibility’ bias can be corrected using distance sampling methods (Buckland et al. 2001). However, if these distances are measured with error, the population density estimators have a substantial bias (Alldredge et al. 2007). When birds are predominantly detected by their audible cues, estimating the distance from the observer is quite difficult in practice.

Detectability is not only a function of the observer but also the focal organism. To be detected, an individual needs to emit some form of a cue. Hence the rate at which cues are emitted and the amount of time the observer is present at the location affect the probability of detection. The higher the cue rate and the longer the survey duration, the higher the probability of detection. The removal sampling method (Farnsworth et al. 2002, Sólymos et al. 2018a) corrects detection errors that occur because of the lack of cues during the finite survey duration. To implement this method, we need to observe the time at which the first cue was emitted by an individual during the survey. Then, that individual is ‘removed’ from further observation, i.e., any further cues emitted by that individual are ignored. This implies that the observer needs to keep track of all individuals while those may be moving around during the survey duration. This

is a difficult task, especially if many species are monitored simultaneously. Furthermore, the removal sampling method assumes that distance from the observer does not affect the detection of the cue, which does not hold in practice.

Other analytical methods require the observer to survey the same location multiple times. Another assumption they make is that the number of birds present at the survey site remains unchanged between surveys, i.e., there is a ‘closed population’. A commonly used such method for estimating population abundance from the counts of individuals or occupancy status at a survey site is the N-mixture method (Madsen and Royle 2023). The N-mixture method assumes that one can visit the survey locations repeatedly, which can be logistically difficult and costly. Furthermore, the ‘closed population’ assumption that the occupancy status or the abundance at every survey location remains the same across repeated visits is well known to be a problematic assumption (e.g., Bayne et al. 2011, Link et al. 2018, Fogarty and Fleishmann 2021).

In this paper, we propose a method that overcomes many of the above shortcomings. Our proposed method requires only a single visit to the survey location and does not require keeping track of individuals, the time at which they emit the first cue, nor the distances of individuals from the observer. The only information we need to collect is either the occupancy status or the count of individuals observed within a specified distance during the specified duration at the survey locations. Despite requiring so little information, we show that our method leads to an estimator of population density that is nearly as good as the estimator that uses exact information on distance to the observer and time of the first detected cue. We refer to our method as Single bin QPAD or SQPAD for short. The acronym QPAD (Sólymos et al. 2013) is based on the equation used to describe the expected value of the observed counts, $E[Y] = DApq$, which is a product of density (D), area (A), and time (p) and distance (q) related detection probabilities. The

QPAD method requires collecting information over multiple time and distance intervals to estimate the p and q detection probabilities. The SQPAD approach relaxes the need for collecting information over multiple time and distance intervals and addresses the non-independence between the two components (p, q) of the detection process.

VARIABLE EFFORT SURVEY DESIGN

The variable effort survey design is required for estimating population density using SQPAD method. The following notation will be used throughout the paper to describe SQPAD methodology. Let $s = 1, 2, \dots, S$ where S is the total number of locations that are surveyed.

- y_s is the occupancy status or abundance at the s -th survey location
- X_s is a vector of habitat covariates that affect density at the time and location of the s -th survey location
- Z_s is a vector of habitat covariates that affect detection at the time and location of the s -th survey location
- t_s is the survey duration at the s -th survey location
- r_s is the radius of the circular buffer around the s -th survey location
- Survey area around the s -th survey location is $A_s = \pi r_s^2$. Let R_{max} be the maximum radius one would survey at any survey location and let $R_{min} > 0$ denote the minimum radius surveyed at any survey location.
- T_{max} denotes the maximum amount of time spent at any survey location. Let $T_{min} > 0$ denote the minimum amount of time one spends at any survey location.

The variable effort survey design consists of the following steps.

Step 1: Choose a location from all the potential survey locations. This could be based on simple random sampling, stratified simple random sampling or any other sampling design one wishes to use.

Step 2: Choose a number from $\{T_{min}, 2T_{min}, \dots, T_{max}\}$ randomly. We denote this by t_s . This is the amount of time one would spend at the chosen location.

Step 3: Choose a number from $\{R_{min}, 2R_{min}, \dots, R_{max}\}$ randomly. We denote this by r_s . This is the survey radius at the chosen location.

Step 4: Note the number of individuals or occupancy status observed at that survey location within the survey radius in Step 3 and during the survey duration in Step 2.

Step 5: Repeat Steps 1-4, until a requisite number of locations are surveyed. We want to emphasize that the survey radius and survey duration are different for each survey location. This is in contrast with the usual practice of having these to be the same for all survey locations. We call this a variable effort survey design. Data collected under the variable effort survey design are $(y_s, X_s, Z_s, t_s, r_s)$, namely, observed abundance or occupancy status, covariates for abundance, covariates for detection, survey duration, and survey radius for each survey. We note that there could be habitat covariates that affect both detection and abundance. Such habitat covariates will be part of both X_s and Z_s vectors. Of course, there may be no habitat covariates and we only have (y_s, t_s, r_s) .

Prior knowledge about the cue rate and density of the species can be used to determine the values of T_{max} and R_{max} . For example, T_{max} maybe small for high cue rate species and vice versa. For multi-species surveys, this may be based on the species with the lowest cue rate. Similarly, for high-density species, one may choose a smaller R_{max} and vice versa. Exact knowledge of these upper limits is not necessary, but expert judgment is useful in designing

surveys that do not spend excessive amounts of time at a survey location. The goal is to ensure that there are sufficient number of locations that are occupied or have non-zero counts, and also to ensure that the upper limits are not so large that occupancy status or counts do not change with increasing time/radius.

STATISTICAL MODELS

We describe two different statistical models, one for count data and another for occupancy data, that will be used to analyze the data collected according to the variable effort survey design described above. These are hierarchical models where the outcome of the process model is fed into the observation model.

Model for Count Data

Hierarchy 1 (Process model): $N_s | X_s, A_s \sim \text{Poisson}(D_s A_s)$, where D_s denotes the density and N_s is the true abundance at the survey location. We assume a log-linear model $\log D_s = X_s \beta$. Notice that N_s are unobserved quantities denoting the true abundance at the time of the survey at the location within the survey area $A_s = r_s^2 \pi$. The Poisson distribution may be replaced by any other discrete distribution such as a Zero-Inflated Poisson (ZIP) distribution or a Negative Binomial distribution.

Hierarchy 2 (Observation model): $Y_s | (N_s = n_s, Z_s, t_s, r_s) \sim \text{Binomial}(n_s, p_s)$, where p_s denotes the probability of detecting an individual at the s -th survey time and location. This assumes that the detections of individuals are independent of each other, probability of detection is the same for all individuals, and that there is no double counting within the survey radius r_s and survey duration t_s .

Model for Occupancy Data

In the field situations, sometimes it is difficult to count the exact number of individuals but an observer can be certain that there was at least one individual present. To model such data, we replace the Poisson distribution with the corresponding censored Poisson distribution, where all non-zero counts are converted to 1's. In this case, a '1' indicates one or more individuals were observed and a '0' indicates no individual was observed. Notice that N_s are unobserved quantities denoting the true occupancy status at these locations.

Hierarchy 1 (Process model): $N_s | X_s, A_s \sim \text{Binomial}(1, \psi_s)$, where ψ_s is the probability of occupancy. We model the probability of occupancy as a function of the density, $\psi_s = 1 - \exp(-D_s A_s)$.

The censored Poisson distribution may be replaced by censored versions of any other discrete distributions such as a Zero-Inflated Poisson (ZIP) distribution or a Negative Binomial distribution. We warn the users that one needs to study the identifiability of the parameters when using a censored version of the ZIP or other zero-inflated distributions.

Hierarchy 2 (Observation model): $Y_s | (N_s = n_s, Z_s, t_s, r_s) \sim \text{Binomial}(n_s, p_s)$, where p_s denotes the probability of detecting that the location is occupied when it is occupied, where $n_s = 0$ or 1. This model can be extended to cases, where the censoring value for N_s is 2, 3 or any other positive integer. This is helpful because individual recognition may be possible when the number of individuals is small, say 2 or 3, but beyond that, it may be difficult. We describe the likelihood functions for general censoring value in the Supplementary Material. The *detect* R package (Sólymos et al. 2025) has a function that allows the user to specify the censoring value.

Modelling Probability of Detection

The key to correcting detection errors using only a single visit is proper modeling of the probability of detection. There are two requirements (Sólymos and Lele 2016): (1) Probability of

205 detection should depend on at least one continuous covariate and this covariate should not affect
 206 the density parameter D , and (2) the functional form of the model for the probability of detection
 207 satisfies the Resource Selection Probability Function (RSPF) condition (Sólymos and Lele
 208 2016). We show that these requirements are naturally satisfied if we appropriately incorporate
 209 the mechanisms of detection. First, detection probability is a function of at least one known
 210 continuous covariate, the survey duration. Furthermore, we show that a reasonable detection
 211 mechanism naturally leads to a model for the probability of detection that satisfies the RSPF
 212 condition.

213 An individual is not detected if it emits no visual or aural cue during the survey duration.
 214 Let the cues be generated according to a renewal process with independent increments. Let the
 215 time between cues be modeled by a positive random variable with a distribution function $F(\cdot)$.
 216 Let $1/\delta$ be the average time between cues. Then, using the theory of renewal processes, it can be
 217 shown that the probability of generating at least one cue in the time interval $(0, t]$ is given by
 218 $\delta \int_0^t (1 - F(y)) dy$. For a Time-Homogeneous Poisson Process (THPP), the time between cues
 219 has an Exponential distribution with rate δ (or, equivalently mean time between cues is $1/\delta$) and
 220 the probability of at least one cue being generated before time t is given by $p(t) = 1 -$
 221 $\exp(-\delta t)$. It is reasonable to assume that if an individual is at a distance 0 from the observer and
 222 emits a cue, it will be perceived with probability 1. Hence the probability that such an individual
 223 will be detected during the interval $(0, t]$ is $p(t) = 1 - \exp(-\delta t)$. Notice that as $t \rightarrow \infty$, the
 224 probability of detection converges to 1 for the THPP model as well as for any renewal process
 225 with independent increments. This says that if we stay at a survey location long enough,
 226 eventually we will detect the species if it is present. Observe that the probability of detection is
 227 not a log-linear function and it strictly increases from 0 to 1. Hence this basic model satisfies the

228 RSPF condition (Lele and Keim 2006, Sólymos and Lele 2016, and the Supplementary
 229 Material). Hence, under this mechanism for detection, one can estimate the density parameter
 230 (D) using a single visit to a survey location.

231 We will now incorporate the effect of distance from the observer on the probability of
 232 detection. It is known that distance affects the perception of the cue: the farther an individual
 233 bird is, the less likely that it will be perceived. If we model the arrival of cues as a renewal
 234 process, the distance acts as a thinning mechanism and reduces the number of cues perceived by
 235 the observer.

236 Thinning of a renewal process results in a decrease in its intensity function or
 237 equivalently, in our case, decrease in the perceived cue rate. With a slight change of notation, let
 238 δ_0 be the innate cue rate. Now we make the perceived cue rate to depend on distance. For
 239 example, one can model the *perceived* cue rate for an individual that is at distance r by
 240 $\delta(r, \varphi) = \delta_0 \exp(-r/\varphi)$. The rate of decrease is controlled by the parameter $\varphi > 0$. Notice that
 241 at distance 0, this is equal to the innate cue rate. As distance increases, the *perceived* cue rate
 242 decreases from δ_0 to 0. A somewhat less drastic change in the perception could be modeled by
 243 $\delta(r, \varphi) = \delta_0 \exp(-(r/\varphi)^2)$. Combining the above results, we can now model the probability of
 244 detecting a bird that is at distance r during a survey duration of t as $p(t, r; \delta_0, \varphi) = 1 -$
 245 $\exp(-\delta_0 \exp(-r/\varphi) t)$. As discussed above, another possible model for the probability of
 246 detection is $p(t, r; \delta_0, \varphi) = 1 - \exp(-\delta_0 \exp(-(r/\varphi)^2) t)$. Notice that this model for
 247 probability of detection has the same form as the well-known complementary log-log link
 248 function used in the generalized linear models.

249 *Note:* We can allow for even more general forms of perception rate by using a model:
 250 $\delta(r; \delta_0, \varphi) = \delta_0 v(r, \varphi)$ for $\delta_0 > 0$ and any function $0 < v(r, \varphi) \leq 1$ for all values of (r, φ) .

Similarly environmental covariates such as time of the day, season, fog or rain, etc. that may affect the innate cue rate can be incorporated by using the model $\delta(r; \delta_0, \varphi, \gamma) = \delta_0 \exp(Z\gamma) v(r, \varphi)$ where Z is the vector environmental covariates that affect the innate cue rate. We note that here the design matrix Z is without the column of 1's as the first column unlike the linear regression design matrix. The interpretation of δ_0 , when centered covariates are included, will be different; it is the cue rate at the average values of the covariate values. Again, this is similar with linear regression where the interpretation of the intercept term is the value of the response at the average values of the covariates. We will not discuss these generalized models and their interpretation further in this paper.

In SQPAD monitoring design, we do not collect data on individual bird location or the time it was observed. We only collect how many individuals were observed or whether there was at least one individual in the survey location. Given this limited data, we cannot compute the probability of detection for an individual bird. As individuals are at different distances from the observer, the probability of detection is not the same for all, which violates the assumption behind the Binomial distribution used in the *Statistical models* section. The resultant distribution is, in fact, a convolution of independent Bernoulli distributions (see the Supplementary Material for details). One can approximate the convolution distribution by a Binomial distribution using the average or median or some such central tendency measure of the individual probabilities (Liu and Quettermous, 2018). We can also use the following approximation of the central tendency measure of the individual probabilities.

Under the assumption that the bird locations are distributed according to the (spatial) Homogeneous Poisson Process (HPP), it is known that the average distance of an event from the centroid of a circle of radius r is $0.67r$. We propose to use the probability of detection computed

at $0.67r$ for approximating the detection parameter of the Binomial distribution. Thus, with the Exponential perception function, we use $p(t, r; \delta_0, \varphi) = 1 - \exp(-\delta_0 \exp(-0.67r/\varphi)t)$ and with the Half Gaussian perception function we use $p(t, r; \delta_0, \varphi) = 1 - \exp(-\delta_0 \exp(-(0.67r/\varphi)^2)t)$. Another possibility, using the mean value theorem of integration, is to use $0.5r$ instead of $0.67r$. In our simulations (see Supplementary Material), these different approximations do not seem to affect the final inference about the population density.

One of the reviewers of this paper pointed out that under the assumption that N is a Poisson random variable, the likelihood function for SQPAD is the same whether we use a Binomial distribution or a convolution of independent Bernoulli random variables as is done in the Supplementary Material. However, we do not know if such equivalence will hold true if N is not a Poisson random variable.

SIMULATION STUDY FOR THE PERFORMANCE OF SQPAD

We now show that the statistical inferences using SQPAD monitoring method are comparable to the case when exact distance and time information for individual birds are available for data scenarios in our simulations. This shows that low requirements for information collection for SQPAD do not substantially affect its statistical efficiency. Furthermore, we show that SQPAD also lends robustness against double counting and errors in distance estimation. In practice, we want a method that is robust and reasonably efficient.

A Mechanistic Model for Simulating Bird Locations in a Study Area

We describe in detail a simple yet plausible individual-based mechanistic model for the location of birds over a landscape. This model can be modified to reflect the behavior of different species of interest. We used the ‘*bSims*’ R package (Sólymos, 2023) to generate such data in R 4.4.2 (R Core Team 2024). The following describes the simulation of bird locations within a single

297 survey location. This can be repeated over several survey locations to mimic the field data. Let
298 us consider the landscape to be a rectangle with sides b_X and b_Y . Let D be the density and $A =$
299 $b_X b_Y$ be the area of the survey location.

300 Step 1: Generate a random number from $Poisson(DA)$. Let this be denoted by N .

301 Step 2: Generate N random numbers from $Unif(-b_X/2, b_X/2)$ for the X -coordinates and
302 another set of N random numbers from $Unif(-b_Y/2, b_Y/2)$ for the Y -coordinates, with
303 coordinate $(0, 0)$ being the center of the rectangle. Then $(X_i, Y_i), i = 1, 2, \dots, N$ are the locations
304 of the N individual birds within the survey location. Note that these X and Y values are not
305 related to the covariates and responses described in the notation. These simply refer to the
306 horizontal and vertical axes.

307 This spatial distribution of the bird locations corresponds to the assumption that bird
308 locations follow a HPP with intensity D . We assume that the observer is at the center of the
309 landscape $(0, 0)$. Notice that birds are located at different distances from the observer.

310 **Mechanism for Generating Cues**

311 It is assumed that every individual bird emits cues over time. We do not differentiate between
312 audible and visual cues. Time between cues is assumed to be independent of when the last cue is
313 emitted. We also assume that the time between cues follows an Exponential distribution with rate
314 parameter δ . Thus, if we monitor an individual bird for a survey duration of, say, T , we can plot
315 the time history of the cues. A time history of cues of hypothetical survey location with five
316 individuals is shown in Figure 1.

317 In statistical parlance, this process is called a Marked Point Process where a ‘Mark’
318 corresponds to the individual identification, and the ‘Points’ are the times at which that
319 individual emitted a cue. We can modify this simple mechanism to incorporate possible

dependence of cue rate on the season, time of the day, and other environmental conditions.

Similarly, we can allow the change in the density from one location to another depending on the environmental covariates.

What Do We Observe?

Unfortunately, in field situations, our observations are not identical to Figure 1. What the observer can observe is the ‘perceived’ version of this ideal situation. The perceived process is shown along the horizontal axis. For example, individual 2 was not detected because it did not emit a cue during the survey duration. On the other hand, individual 4 emitted a cue but was not perceived by the observer, possibly due to the distance. Individuals 1 and 3 were perceived at the time of the first cue they emitted whereas individual 5 was detected but at the second cue, not the first one. We missed the first cue from individual 5 possibly because of the distance. The third cue from individual 5 is ignored because we have already counted the individual 5 and hence ‘removed’ from consideration. Overall, we detected 3 individuals from the 5 individuals that were present.

Hence the perceived process is quite different from the actual process because of the lack of cues and effect of distance from the observer. The challenge is to use the perceived process to infer the underlying true process, in particular, the bird density D .

Now let us mimic the data collection process according to various monitoring methods.

1) The complete-information method, based only on the solid dots, would declare that there are 3 individuals and notes both the distance and time information, namely, $(t_1, d_1), (t_3, d_3), (t_5, d_5)$.

2) Distance sampling notes that 3 individuals were observed at distances d_1, d_3 , and d_5 , respectively. It ignores the time of the cue. These are the projections of the solid dots on

the distance axis (Figure 1). Notice that we keep the information on the identity of the individuals. In practice, instead of the exact distances, one may only note the distance interval in which the observation belonged to. There should be at least two pre-defined distance intervals.

- 3) Removal sampling also notes that 3 individuals were observed but collects information on the time of the first ‘perceived’ cue only at t_1 , t_3 , and t_5 , respectively. These are the projections of the solid dots on the time axis (Figure 1). Notice that we keep the information on the identity of the individuals. In practice, instead of the exact times, one may only note the time interval in which the observation belonged to. There should be at least two pre-defined time intervals.

- 4) SQPAD declares that 3 individuals were observed during the survey duration $(0, t)$ and within a spatial buffer of radius r . It does not collect information about individual cue times or distance from the observer.

We collect similar information at all survey locations. This is the kind of dataset we usually have for estimating the bird density. Obviously, this is a description of a simple situation that does not involve effects of covariates on the density, cue rate or visibility. Those complications, however, can be accommodated.

Now we compare statistical properties of the estimators of bird density using distance sampling, removal sampling and SQPAD with the estimator that uses the complete information. The likelihood function for the complete-information data is described in the Supplementary Material. We consider four different situations: low density ($D = 0.5$) and low cue rate ($\delta = 0.25/\text{min}$), low density ($D = 0.5$) and high cue rate ($\delta = 0.5/\text{min}$), high density ($D = 1$) and low cue rate ($\delta = 0.25/\text{min}$), and high density ($D = 1$) and high cue rate ($\delta = 0.5/\text{min}$). The

parameter $\varphi = 100$ m was the same for all the cases. The number of survey locations (S) was 200. We generated 1000 independent datasets and we sampled from it without replacement.

Recall that monitored data are different than the complete-information data. We used different-sized spatial buffers and different survey durations for each survey location. The maximum time duration (T_{max}) was 10 minutes and the maximum spatial radius (R_{max}) was 200 meters. The minimum amount of time spent (T_{min}) was 2 minutes. Thus, duration t_s for the s -th location was one of $\{2, 4, 6, 8, 10\}$. The minimum spatial radius (R_{min}) was 20 meters and spatial buffer radius, r_s for the s -th location was one of $\{20, 40, 60, \dots, 200\}$.

The detection probability was based on the Half-Gaussian distance effect model with $\varphi = 100$ m. Thus, the probability of detecting an individual at a distance r is $p(t, r; \delta_0) = 1 - \exp(-\delta_0 \exp(-(r/\varphi)^2)t)$. The Half-Gaussian model has a shoulder suggesting a slower drop in the perception of the cues near the observer than for the Exponential model. This model seems to represent the reality better than the Exponential model. These data can be analyzed using SQPAD method as well as all the traditional methods of Distance sampling, Removal sampling, and the QPAD.

For the QPAD analysis (Sólymos et al. 2013) as well as distance sampling (QAD) and removal sampling (PAD), we constructed the distance bins for a particular survey location by dividing its spatial radius r_s equally in 3 parts and time bins by dividing its survey duration t_s equally in 3 parts. To apply QPAD, we need to bin the distance-time data as described in the *Field data analysis* section. Bin sizes are different at different survey locations because the survey duration and survey buffer are different at every location. We used multinomial distance sampling and removal sampling approaches as reviewed in Sólymos et al. (2013). This analysis was conducted using the ‘detect’ R package (Sólymos et al. 2025).

The Naïve method, that completely ignores the detection error, corresponds to using a Poisson regression model with offset for the area of the survey location. Notice that the areas are different at each survey location. This method is included to illustrate the importance of incorporating detection error in the analysis.

Simulation Results

We start with the situation when distances are measured without error and individuals are tracked perfectly and individuals do not move during the duration of the survey (Figure 2). As expected, the Naïve method is negatively biased. The complete-information likelihood, the convolution likelihood estimator, is nearly unbiased and its sampling distribution is also nearly symmetric. In the *Statistical models* section we discussed two versions of SQPAD; one based on the counts and one based on the occupancy status. Despite not collecting complete information, the performance of SQPAD estimator comes reasonably close to the complete-information estimator, both in terms of bias, variance, and symmetry. SQPAD estimator based on occupancy (SQPAD-O) is surprisingly comparable to SQPAD estimator based on counts (SQPAD-C). On the other hand, distance sampling (denoted by QAD), removal sampling (denoted by PAD), and QPAD have substantial mean biases. The sampling distributions of these estimators are also highly skewed, generally leading to density estimators that are substantially larger than the true density. This figure illustrates that SQPAD works and is comparable to the complete-information estimator for the data scenarios in our simulations. We also report results on the estimation of cue rate δ and distance effect parameter φ , including biases and variances in a tabular form as part of the Supplementary Material.

To study the effect of errors in distance estimation and double counting on these estimators, we introduce these errors in the data generation mechanism. See Sólýmos (2023) or

the description of the ‘*bSims*’ R package for details on how to introduce these errors in the data generation mechanism. It is known (Aldredge et al. 2007) that errors in distance estimation tend to shorten the true distances (Aldredge et al. 2007). To mimic these errors, we simply multiplied the true distances by 0.8 to get the ‘observed’ distances. We introduced double counting by putting a probability of observing multiple cues emitted by the same individual as multiple individuals at 0.2. This leads to expected counts that are 1.2 times higher than the true counts. For these two scenarios, we also introduced territorial movement for the individuals according to a bivariate Normal kernel (Sólymos 2023). Territorial movement adds variability to the distances and is also a mechanism for double counting of individuals. Figures 3 and 4 summarize these results. As expected, all estimators are biased. However, SQPAD estimators, occupancy-based and count-based, are the least biased and have smaller variability among all the estimators. This illustrates their robustness. A detailed simulation study of the coverage properties of the confidence intervals based on SQPAD estimators needs to be conducted.

In our simulations with double counting error (Figure 4), the occupancy based SQPAD estimator was more biased than the count-based approaches (convolution likelihood and SQPAD-C) when density and cue rates were both high. This is possibly caused by having too few 0’s (10–24%) in the simulated datasets. It is generally known that occupancy-based methods are less accurate when counts are high and the frequency of 0 values is low.

The QPAD estimator proposed by Sólymos et al. (2013) uses the same information as the complete-information convolution likelihood estimator. However, it uses a somewhat untenable assumption that the probability of detection can be written as a product of two probabilities: one that depends only on the distance and the other that only depends on the time. This is what is termed as the ‘independence model’ (Sólymos et al. 2013, Martin-Schwarze et al. 2021). This

assumption allows the estimation of the cue rate and visibility bias in a simple fashion from field data collected under many different protocols (Edwards et al. 2023). However, if this assumption is not satisfied, as is the case in our simulations, the QPAD estimator of density is biased. This bias is evident in Figures 2–4. We provide information on the behavior of the estimates of cue rate and distance effect parameters in the Supplemental Material. As is the case for density estimates, SQPAD estimates are better than QPAD, distance sampling, and removal sampling estimates.

AN EXAMPLE FIELD DATA ANALYSIS

We reanalyzed data from Van Wilgenburg et al. (2017) using SQPAD. We present results only for 4 of the species that were monitored: *Spizella passerina* (Chipping Sparrow), *Junco hyemalis* (Dark-eyed Juco), *Seiurus aurocapilla* (Ovenbird), and *Catharus ustulatus* (Swainson’s Thrush). These species represent different average densities and cue rates based on previous QPAD estimates (Van Wilgenburg et al. 2017). Code for SQPAD analysis and results for the other 8 species can be found in the Supplemental Material.

The data consisted of 351 point count surveys performed by human observers. We used latitude (centered and scaled by its standard deviation) as a covariate for density (X) and time since local sunrise (hours since sunrise divided by 24) and its quadratic term as covariates for the cue rate (Z). For each survey, distances to the individuals were reported to belong to one of two bins; less than 50 and 50–100 m. Similarly, times to the first cue were reported to belong to one of three bins: less than 3 minutes, 3–5, and 5–10 minutes.

A hypothetical but typical datum corresponding to a single survey is a two-way table (Figure 5A). At this survey location, we observed 2 distinct individuals during the first 3 min and both were within 50 m of the observer. These two individuals are excluded from further

consideration. Other cells can be interpreted similarly. We observed a total of 5 individuals within 100 m of the observer location and during the 10-min duration.

This field data was not collected according to the variable effort survey design. However, one can turn these data into data observed under the variable effort survey design. For each survey location, one can randomly select a number from (50,100) and another number from (3,5,10). Suppose for the above survey location, random numbers were 50 and 5, respectively. Then we can replace the above two-way table with a vector $(y_s, r_s, t_s) = (3, 50, 5)$ as the observed data for that survey location.

The scheme described above of turning a typical distance-time binned dataset into the variable effort survey design dataset, however, loses information because we use one randomly selected subset from all possible subsets. The number of possible subsets is equal to the number of distance bins multiplied by the number of time bins. In the case above, there are 6 possible subsets. We can estimate density D using any one of these subsets. Alternatively, we can estimate D using all 6 subsets and averaging those estimates to get a better estimate that uses full information in the data. The method of composite likelihood (Lindsay 1988, Lele 2006) suggests that instead of averaging the estimates, one can average the corresponding log-likelihoods and maximize the average log-likelihood with respect to the parameters.

Let us consider the hypothetical two-way table above. There are 6 possible sub-tables that we can consider for analysis using SQPAD (Figure 5B). Of course, these 6 data points corresponding to a single survey location are not independent. The estimators obtained by maximizing the composite likelihood are known to be consistent, i.e., they converge to the true parameter value as sample size increases. However, the confidence intervals for these estimators cannot be based on the inverse of the Hessian matrix as is done for the maximum likelihood

481 estimators. Instead, the non-parametric bootstrap can be used to compute the correct confidence
482 intervals.

483 The bootstrapping is done as follows. Suppose there are S survey locations.

484 Step 1: Obtain a simple random sample with replacement of size S from the survey location
485 indices $(1, 2, \dots, S)$.

486 Step 2: For the survey locations with indices obtained in Step 1, transform the two-way tables
487 into corresponding data vectors as described above.

488 Step 3: Analyze the resultant data using SQPAD method to get the density estimate.

489 Step 4: Repeat steps 1–3, B number of times.

490 The bootstrap estimate of the sampling distribution is simply the histogram of these B bootstrap
491 estimates. This is used to compute the confidence interval for the density estimate. Nevertheless,
492 we bootstrap the survey locations and not the constructed variable effort survey dataset.

493 The distance-time binned data described above is commonly collected in many
494 monitoring studies. Following the above scheme, these datasets can be turned into variable effort
495 survey design datasets and analyzed using SQPAD method. Thus, all these previous datasets can
496 be fruitfully reanalyzed.

497 The results of SQPAD analysis along with the QPAD analysis and the naïve analysis of
498 the field dataset described earlier are presented in Figure 6. We used the Half-Gaussian visibility
499 function for both analyses. This reanalysis shows that the density estimates using QPAD are
500 substantially higher than those indicated by SQPAD analysis (Figure 6). This positive bias in the
501 QPAD density estimates was also observed in the simulations. Notice that, for these data,
502 SQPAD and naïve estimates of density are quite similar. We suggest that in point count data
503 analysis, researchers present naïve analysis along with an analysis that corrects for detection

error. A meta-analysis of these data will help us understand if the detection error is, in fact, an important biasing factor in practice. . SQPAD estimates of the cue rates for the 12 species estimates also look biologically plausible (Supplemental Material).

While it would have been ideal to illustrate SQPAD on data that were collected under our proposed variable effort survey design, unfortunately, such data are currently not available. Hence, we decided to use the available prior data that were collected in multiple time-distance bin format. We developed the composite likelihood method to illustrate SQPAD analysis and also to assure researchers that their previous datasets can be analyzed using SQPAD and inherit its good properties. If there were issues with distance estimation or recognition of the individuals, SQPAD analysis will be robust against such potential lacunas.

MODEL SELECTION, MODEL DIAGNOSTICS AND GOODNESS OF FIT TESTING

Anytime a statistical model is fit to a dataset, we should explore the possibility of alternative models and some measures of goodness of fit. In the following, we discuss three possible approaches in the context of SQPAD. We will study performance of these in a separate paper.

- a) *Model selection*: As we have discussed earlier, there are various possible models for the distance effect on detection as well as the effect of the covariates on the bird density D . Similarly, one can possibly consider different arrival process models for the cues. We suggest the use of the well-established Bayesian (Schwartz) Information Criterion (BIC/SIC) (Burnham and Anderson 2002), as a general tool towards this goal. Although, given their variability, large datasets are required for these criteria to effectively distinguish among models (Dennis et al. 2019, Taper et al. 2022). This model-selection process can be used only for: (1) selection of covariates that affect bird density D , (2) selection of different forms of distance-detection functions such as Half Gaussian vs.

Exponential, and (3) selection of covariates such as time of day or season that may affect cue rate.

We suggest that the following steps should be taken while fitting the detection-error model:

Step 1: Fit the ‘no detection error’ model as the basic (or, null) model. For SQPAD, this is simply a Poisson regression estimation of the density parameter D with no adjustment for the detection error. One can also conduct standard residual analysis, Chi-square measure of goodness of fit etc. along with leave-one-out type cross validation prediction error analysis.

Step 2: Include detection error model with only the time effect. That is, use the detection probability model that includes only the survey duration as the covariate. Compare the basic (or, null) model fit to the model that uses detection error due to survey duration and arrival process for the cues.

Step 3: Include the distance along with the survey duration in the detection probability model. Choose the form of the ‘visibility bias’ or ‘distance effect’ model. Once the distance effect is included in the model, one can use information criterion or cross validation to compare and choose between the null model, time effect only model and time-distance effect model.

b) *Goodness of fit diagnostics:* A traditional goodness of fit test, such as based on Pearson Chi-square, can be used. However, such tests have their supporters and detractors. As no model represents reality perfectly, given sufficient data, one can always reject any model. Also a model can be wrong in many directions and these tests cannot have good power in every such direction. We believe that one can compare the appropriateness of two models

but model adequacy of a single model is a tricky and essentially an unsolved problem because what is an adequate fit is a decision problem; a particular Chi-square value might be reasonable for some kinds of decisions but not for others.

c) *Model diagnostics*: The model selection procedure and goodness of fit tests, being omnibus tests, cannot diagnose validity of individual components of the model for density parameter and model for detection error separately. For example, to judge the appropriateness of the time-distance effect on the detection, one needs to have a complete record for every individual: the time of all the cues and the distance, at which they emitted (e.g., Martin-Schwarz et al. 2021). This kind of information, without any errors in data collection, is nearly impossible to get in practice. This is particularly true when the cues are aural rather than visual. Hence, we doubt the practical utility of such diagnostic tests in bird monitoring.

Furthermore, in practice, distance and first arrival times data are usually collected in 2 or 3 intervals. Given such crude resolution data, the power to discriminate between different forms of distance effect models is likely to be quite moderate. On the other hand, although SQPAD samples only one bin at any location, by varying the time and distance bin sizes at a finer scale from location to location, we expect SQPAD to be more sensitive to model differences than QPAD and distance sampling. The sensitivity to differences in the detection models under these different protocols needs to be studied further.

Some Caveats

SQPAD model is a hierarchical model – a model that minimally consists of two components: a process model and an observation model (Lele et al. 2007) – and an inherent characteristic of any hierarchical model is that one cannot separately check the validity of the individual components

of the model given only the observed data. Thus users of SQPAD model should assess whether these assumptions are reasonable for their specific system.

Assumption 1: SQPAD model assumes that the locations and numbers of birds within the surveyed area are determined by an HPP. This implies that the counts within a buffer is Poisson random variable with mean proportional to the area of the buffer. This assumption cannot be justified using the observed data alone. This assumption can be replaced using some other discrete distribution such as zero inflated Poisson distribution or Negative Binomial distribution, if this change is appropriate to match known or presumed characteristics of the underlying spatial point process. However, again such an assumption cannot be justified using the observed data alone.

Assumption 2: The basic formulation of SQPAD also assumes that cues emitted by individual birds arise from a time HPP and that individuals are emanating cues independently of each other and at the same rate. Again, potentially one can specify some other counting process model for the arrival of cues if warranted from external knowledge of the system. However, the assumptions about the cue process cannot be justified using the observed data alone.

The inability to test these two model assumptions based only on the data used to fit a model is not unique to SQPAD; other methods also have assumptions that are not testable using the observed data. For example, Distance sampling method assumes that locations of birds are distributed in space according to a spatial HPP, and that the probability of detection at 0 distance is 1. Neither of these assumptions can be justified using only the data at hand. They are justified based on expert judgement alone.

Removal sampling assumes that the cues from individual birds arrive according to a time HPP and that birds emit cues independently of each other and with the same rate. Neither of

these assumptions can be justified using only the first arrival times. An Exponential distribution of first arrival times does not necessarily imply that the entire arrival process is a time HPP, nor does it imply that all individuals have the same cue rate nor that they emit cues independently of each other.

Being a combination of distance sampling and removal sampling, evaluating the validity of QPAD inherits the same challenges as these two methods. Similarly, Occupancy and N-mixture models assume population closure and independence among visits but cannot justify these assumptions based only on the observed data.

In summary, SQPAD method is not unique among the methods for estimating density in the presence of detection error in requiring assumptions that cannot be evaluated with the data used to fit these models. They all require assumptions that cannot be justified using the data alone.

In fact, if we are only interested in learning about the bird density and not the cue rate, SQPAD relaxes the assumption that the distance effect function be 1 at distance 0. Only assumption we need is that the detection probability be a function of the survey duration such that it is 0 for survey duration 0 and strictly increases to 1 as survey duration increases (see Supplementary Material and Sólymos and Lele 2016). From the scientific perspective, we believe that it is the density, and not the behavior of the birds nor the observation process, that is generally of interest. We look at the detection error model as a ‘statistical adjustment’, based on some understanding of the detection mechanism. Generally, we do not recommend interpreting the estimated cue rate or distance effect parameters as a representation of some physical reality.

DISCUSSION

Surveying bird populations is labor-intensive, logistically difficult, and time-constrained due to inherent seasonality. Hence, we need to develop statistical methods that require data that can be reliably and cost-effectively collected in the field. Effects of errors in data collection, such as distance estimation error (Alldredge et al. 2007) on density estimation are well-known. Our simulations indicate that double counting also leads to bias in density estimation. Similarly, the independence assumption used in QPAD leads to significant biases (Martin-Schwarze et al. 2021). We have shown that SQPAD is a viable alternative to these traditional methods. First and foremost, the data requirements for SQPAD are much less stringent than these traditional methods. Not having to keep track of individuals and not having to estimate the exact distances can simplify the mental effort needed by the observer and would likely to lead to more reliable counts.

Our simulations show that, if the detection model is correctly specified, one can account for the effect of the cue rate and visibility bias on detection without requiring the collection of detailed spatio-temporal information. We do assume, however, that the observer can reliably decipher if an individual is inside the specified spatial buffer. Individuals that are at the edge of the boundary could be potentially misclassified, leading to some bias. Our simulations show that SQPAD inferences are robust against double counting and errors in distance estimation. We also showed that previous datasets collected under a commonly used protocol that is different than the variable effort survey design can also be analyzed using SQPAD method.

We emphasize again that SQPAD and the variable effort survey design remove the need for the difficult task of keeping track of individuals and estimating the distance to a fast-moving individual that is perceived only by an audible cue. Our own experience as well as the experience of our field ecologist colleagues suggests that information about the occupancy status or counts

is far more reliably collected than information on the auxiliary variables, such as individual identification and distance. Reducing the burden to collect these auxiliary variables is also likely to lead to more reliable counts or occupancy status. Such benefits can compound when conducting multi-species point surveys.

For this method to work, we need two assumptions: (1) There is at least one covariate that affects the detection probability but not the density D ; (2) The model for the probability of detection is such that it is a monotonically increasing function from 0 to 1. This is the RSPF condition (Lele and Keim 2006, Sólymos and Lele 2016). We argue that both these conditions are satisfied by SQPAD.

Survey duration is a covariate that is universally available in bird monitoring programs and affects only detection but not density. Moreover, if the survey duration is 0, the probability of detection is 0 and as the survey duration increases, the probability of detection increases. That is, we assume that as survey duration increases, if an individual is present at the location, eventually it will be detected with probability 1. As we have shown, a reasonable mechanism for detection, that also underlies the commonly used methods of removal sampling and QPAD, leads to such a model for the probability of detection. We have added no new assumptions beyond what is assumed in these current methods. The main change that SQPAD method introduces is the variation in the survey duration and survey radius from location to location.

If the variation in survey duration and survey buffer is low or if they are identical from location to location, this method will not work. This is consistent with the result that higher variation in covariate values leads to better regression model estimation (Neter et al. 1983). This was also observed in the context of regression models used in occupancy studies (von Hirschheydet et al. 2023). Misclassification of individuals as being inside the buffer when they

are outside and vice versa may also lead to some bias. However, simulations that include the errors in distance estimation indicate that such misclassification may not have a substantial impact.

We derived the detection probability model based on a specific mechanistic model for the arrival and perception of cues. Although not tested using simulations, we believe, mathematically, we only need to have a detection probability model to be a function of the survey duration (time) such that, with all other covariates including distance held constant, such that the probability of detection strictly increases from 0 to 1 as the duration increases from 0 to ∞ . Any such function along with a spatial homogeneous Poisson point process for the location of individuals guarantees identifiability of the density parameter D (Sólymos and Lele 2016).

When using QPAD, we collect data in multiple time and distance bins at the same survey location. Similar to the results in Lele et al. (2012), what we have shown is that, without any additional assumptions, one can estimate bird density even if one collects data in a single time and distance bin, as long as the bin sizes vary from location to location. The *within* location changes in bin sizes for QPAD is replaced by changes in bin sizes *between* survey locations in SQPAD. This simple modification in data collection protocol allows us to collect data in a simpler fashion without losing any inferential properties of QPAD. We have also removed the troublesome time-distance independence assumption required by QPAD.

As information from Autonomous Recording Units (ARUs) becomes more readily available (Shonfield and Bayne 2017), there is a growing need for analytical methods that work with the characteristics of data from ARUs. ARU datasets typically do not have information on individual distances and identification of individuals is also difficult or impossible, making the use of traditional methods, such as Distance sampling or Removal sampling difficult. On the

other hand, with ARU data, the physical radius r for the spatial buffer can be reasonably replaced by the sound pressure (Yip et al. 2020) and they are also more reliable to get occupancy data than to get the counts. Hence SQPAD-O is well equipped to deal with the ARU data. Like the QPAD, it is also well-equipped to deal with data that are collected under differing protocols between different monitoring programs. In recent years, citizen science data on the occurrence of bird species has become commonly available. Most avid observers can collect occupancy data and if the individual they heard or saw was within a specified buffer. Hence, we suspect SQPAD may be able to deal with such data as well. We will be studying these extensions in a future publication.

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Figure 1. The observation process for a 10-min 100-m radius hypothetical point count survey location where there are five individuals within the survey area. Circles indicate the time and distance of bird individuals giving a cue, vertical arrows pointing towards the observer indicate cues perceived by the observer, and filled dots are cues perceived first for each individual. This figure depicts an “ideal” scenario, where individuals do not move (flat lines) during the survey, there is no distance estimation error and double counting, as a result, 4 cues are detected that belong to 3 individuals. The horizontal axis is the time axis and the vertical axis is the distance axis. A cue emitted from an individual is denoted by a circle, filled (cue detected) and hollow (cue not detected). Each individual’s cue history can be deciphered along with the distance from this figure.

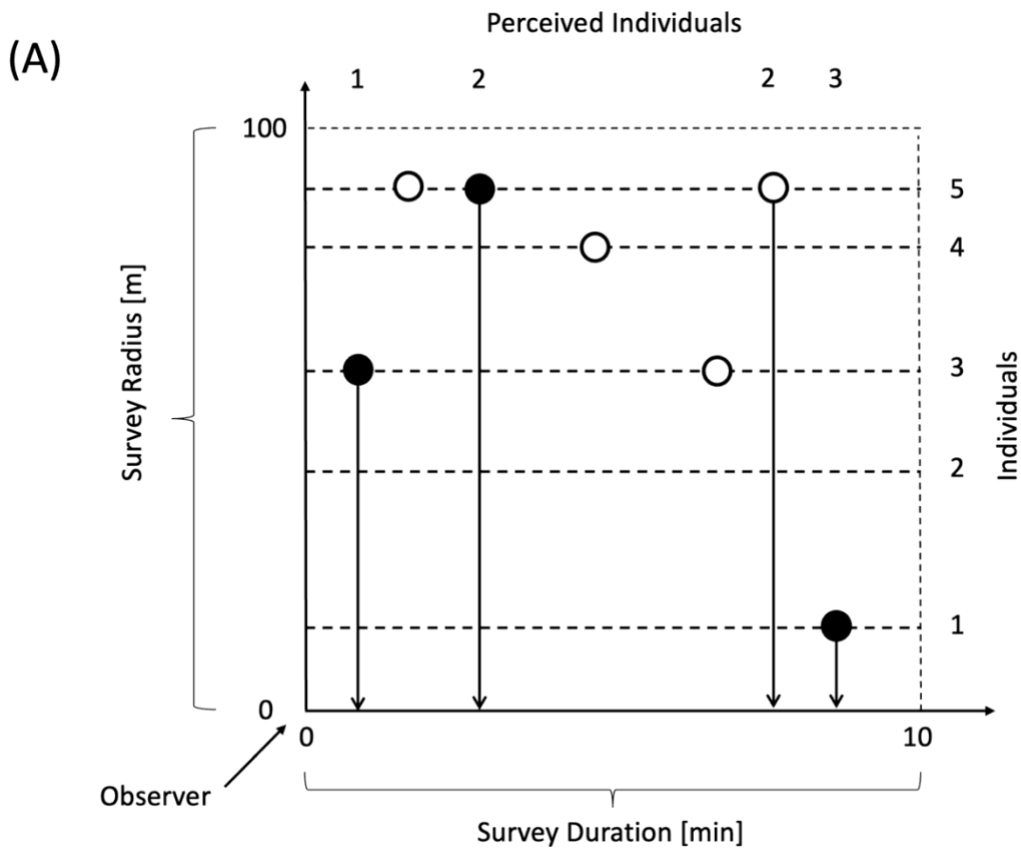


Figure 2. Performance of various estimators (Convolution, SQPAD-C, SQPAD-O, QPAD, Distance sampling [QAD], Removal sampling [PAD], and Naïve) of density D . We subtract the true value of D from the estimates to represent the bias. The dotted line corresponds to zero bias. We use the Half Gaussian model for the distance effect on perception with parameter $\varphi = 100$ m. Number of survey locations is 200 and number of simulations is 2000. We used the 0.67 r approximation to the distance effect. Boxes and thick lines across the boxes indicate 25th, 50th and 75th percentiles, whiskers extend to 1.5 times the inter-quartile range, open circles beyond the whiskers are outliers, x symbols indicate the mean.

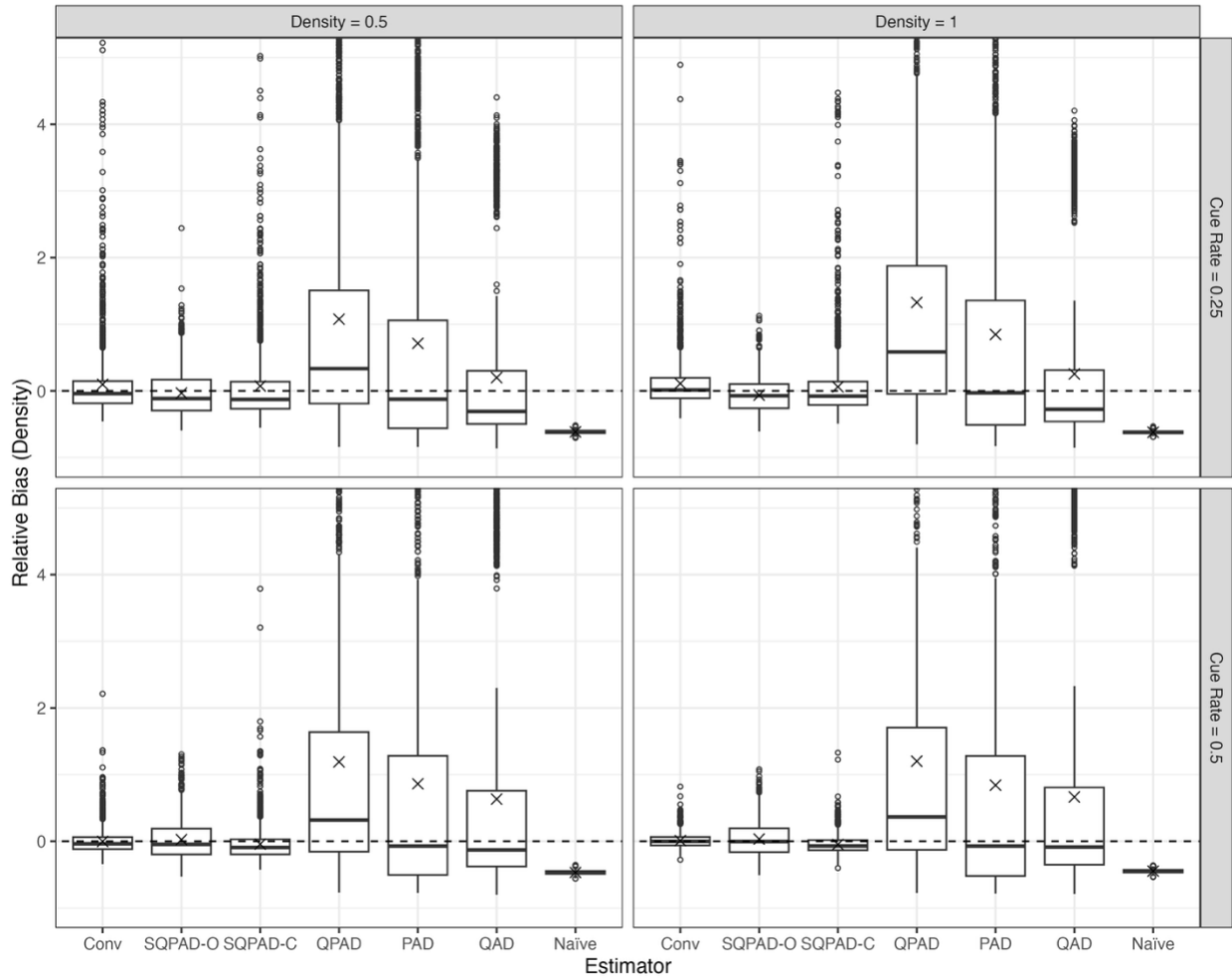


Figure 3. Performance of various estimators (Convolution ML, SQPAD-C, SQPAD-O, QPAD, Distance sampling [QAD], Removal sampling [PAD], and Naïve) of density D when distances are measured with error (20% negative bias). We subtract the true value of D from the estimates to represent the bias. The dotted line corresponds to zero bias. We use the Half Gaussian model for the distance effect on perception with parameter $\varphi = 100$ m. Number of survey locations is 200 and number of simulations is 2000. See Figure 2 legend for the explanation of box and whiskers statistics.

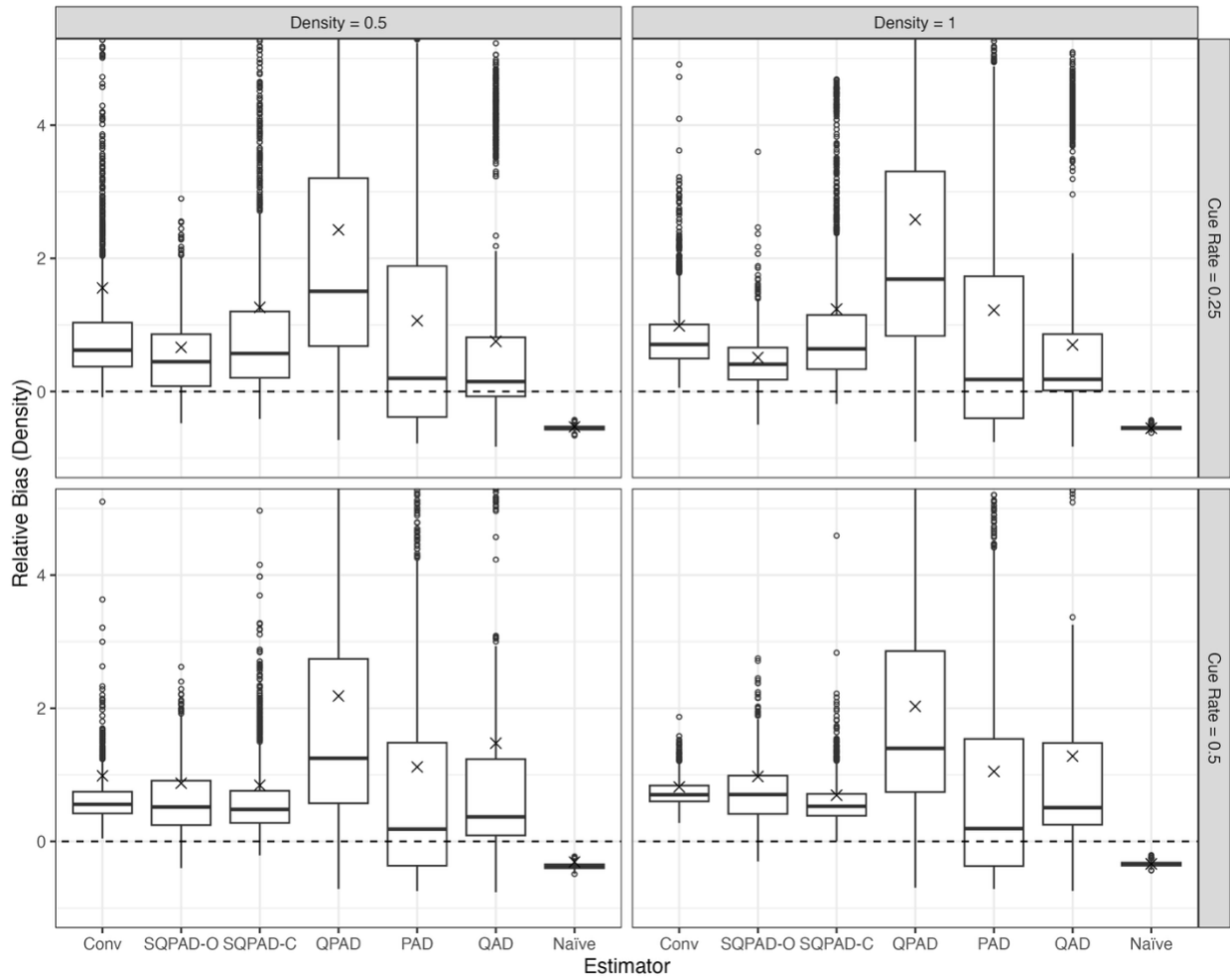


Figure 4. Performance of various estimators (Convolution ML, SQPAD-C, SQPAD-O, QPAD, Distance sampling [QAD], Removal sampling [PAD], and Naïve) of density D when there was double counting (20% positive bias). We subtract the true value of D from the estimates to represent the bias. The dotted line corresponds to zero bias. We use the Half Gaussian model for the distance effect on perception with parameter $\varphi = 100$ m. Number of survey locations is 200 and number of simulations is 2000. See Figure 2 legend for the explanation of box and whiskers statistics.

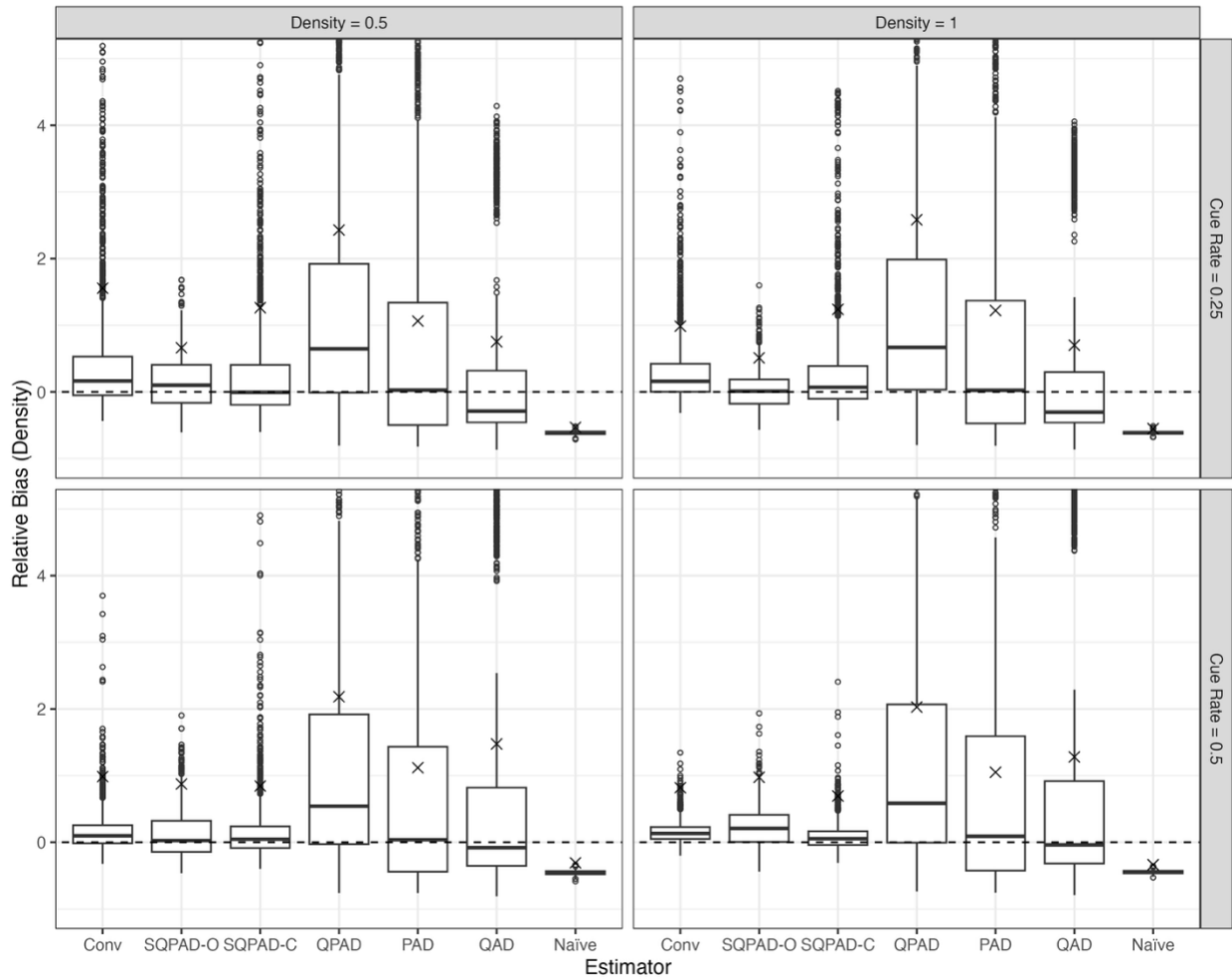


Figure 5. Count of individuals' first detections by time and distance intervals for a hypothetical survey under the combination of removal and distance sampling required for multiple-bin QPAD (A). The same counts from the same survey are organized according to the cumulative time and distance intervals and the corresponding counts as required for the composite likelihood SQPAD (B).

(A)

Time/Distance	0-50 m	50-100 m
0-3 min	2	0
3-5 min	1	1
5-10 min	0	1

(B)

Time	Distance	Count
0-3 min	0-50 m	2
0-3 min	0-100 m	2
0-5 min	0-50 m	3
0-5 min	0-100 m	4
0-10 min	0-50 m	3
0-10 min	0-100 m	5

Figure 6. Results from the reanalysis of field data for 4 species, *Catharus ustulatus*, *Junco hyemalis*, *Seiurus aurocapilla*, and *Spizella passerina*, using SQPAD-C, SQPAD-O, QPAD and Naïve estimators. SQPAD estimates are based on the composite likelihood approach. SQPAD estimates are uniformly lower than the corresponding QPAD estimates. We used the Half Gaussian model for the distance effect on perception.

